

Statistical and Machine Learning

Linear models

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Question 1 – Intro to Linear Models

Suppose that we observe the response Y_i , $i = 1, \dots, n$ with $n = 9$, along with numeric variables w_i and z_i , and a factor variable g_i that contains labels partitioning Y_i into three groups. For convenience, let $g_1 = g_2 = g_3 = 1$, $g_4 = g_5 = g_6 = 2$ and $g_7 = g_8 = g_9 = 3$ (we can always arrange the index so that this holds). We further assume that the following model is considered to be appropriate for the data:

$$Y_i = \beta_0 + \mu_{g_i} + \beta_1 w_i + \beta_2 z_i + \beta_3 z_i^2 + \beta_4 w_i z_i + \varepsilon_i, \quad i = 1, \dots, n.$$

Then, we may re-write the model as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ by writing explicitly each element of $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}_0$, $\boldsymbol{\varepsilon}$ by following these rules:

- Each factor level / group is assigned an indicator variable corresponding to a column in $\mathbf{X}$, such that:
 - $x_{ij} = 1$ for $j = 2, 3, 4$ indicates membership of the i -th observation to the $(j - 1)$ -th group
 - $x_{ij} \in \{0, 1\}$ and $\sum_{j=2}^4 x_{ij} = 1$
- Variables w_i and z_i are included in the model non-linearly.

a) Fill in the first three and the last element of the model in the matrix form below.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon},$$
$$\begin{bmatrix} Y_1 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ Y_9 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ \cdot \end{bmatrix}.$$

b) Answer the following questions:

- Is the model linear in the parameters? How can you conclude that?
 - Is the model linear in errors? How can you conclude that?
 - Is the model linear in regressors/features? How can you conclude that?
 - How many parameters do we need to estimate?
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Question 2 – Properties of the OLS estimator

Besides having a nice geometrical interpretation, the least squares estimator of β_0 has an *analytical expression with respect to $\mathbf{Y}$ and $\mathbf{X}$* . We can re-write the residual sum of squares as

$$S(\beta) = \|\mathbf{Y} - \mathbf{X}\beta\|^2 = (\mathbf{Y} - \mathbf{X}\beta)^\top (\mathbf{Y} - \mathbf{X}\beta)$$

a) Compute the minimizer $\hat{\beta}$ of function $S(\beta)$ and show that it is given by $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

Hint: use in your derivations the following properties:

$$\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{a} = \mathbf{a} \quad \text{and} \quad \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{A} \mathbf{x} = 2\mathbf{A} \mathbf{x}.$$

b) Assuming that $\varepsilon \sim (\mathbf{0}, \sigma^2 \mathbf{I}_n)$, which yields $\mathbf{Y} \sim (\mathbf{X}\beta_0, \sigma^2 \mathbf{I}_n)$, prove that:

- i) $\hat{\beta}$ is unbiased, i.e., $\mathbb{E}(\hat{\beta}) = \beta_0$,
- ii) covariance of $\hat{\beta}$ is given by $\text{cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$.
- iii) Now, additionally, assume that $\varepsilon \sim \mathcal{N}_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ which yields $\mathbf{Y} \sim \mathcal{N}_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_n)$. Given that, why can you conclude that $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is also normally distributed?

Question 3 – Properties of the Hat Matrix $\mathbf{H}$

Let $\mathbf{H} := \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ be the hat matrix defined in the course notes. Show that:

- (a) the trace of $\mathbf{H}$ is the number of identifiable parameters in the model, $\text{tr}(\mathbf{H}) = p$.

Hint: for any real matrix $\mathbf{A}$ it is always true that $\text{tr}(\mathbf{A}^\top \mathbf{A}) = \text{tr}(\mathbf{A} \mathbf{A}^\top)$.

- (b) $\mathbf{H}$ is *idempotent*: $\mathbf{H}\mathbf{H} = \mathbf{H}$,
 - (c) $\mathbf{H}\mathbf{X} = \mathbf{X}$ (i.e. $\mathbf{H}$ is a *projection* matrix onto the space spanned by the columns of $\mathbf{X}$, and hence $\mathbf{H}\mathbf{X} = \mathbf{X}$).
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Question 4 – Properties of Raw Residuals

Let $\mathbf{Y}$ a response vector, $\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\beta}$ and let $\mathbf{H}$ be the hat matrix from Question 3 above.

- (a) define raw residuals in terms of $\mathbf{Y}$ and $\mathbf{H}$.
 - (b) compute $\mathbb{E}(\mathbf{e})$
 - (c) compute $\text{cov}(\mathbf{e})$
 - (d) Assume $\mathbf{Y} \sim \mathcal{N}_n(\mathbf{X}\beta_0, \sigma^2 \mathbf{I}_n)$. What is the distribution of $\mathbf{e}$? Why is so?
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Question 5 – Influential Observations

- (a) What is the difference between an **outlier** and an **influential observation** in a regression context?
 - (b) What does the leverage h_{ii} measure, and what is the commonly used threshold for declaring an observation as high leverage?
 - (c) Cook's distance D_i combines two sources of information. Name them and explain why both are necessary.
 - (d) A practitioner finds Cook's distance $D_i = 0.8$ for a single data point. What action, if any, would you recommend?
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Question 6 — OLS as Best Linear Approximation

- (a) Write down the population-level minimisation problem that defines the OLS parameter β_0 , without assuming the true model is linear.
 - (b) Suppose the true relationship is $Y = f(\mathbf{X}) + \varepsilon$. Write Y in the form $Y = \mathbf{X}^\top \beta + r(\mathbf{X}) + \varepsilon$. What does $r(\mathbf{X})$ represent, and what orthogonality condition must it satisfy?
 - (c) Why does the moment condition $\mathbb{E}\{X(Y - \mathbf{X}^\top \beta)\} = 0$ make OLS estimators meaningful even under misspecification?
 - (d) Why is the statement “OLS only works under linearity” too strong?
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Question 7 — Logistic Regression: Concepts

- (a) Give three reasons why a standard linear regression model is inappropriate for $Y \in \{0, 1\}$.
 - (b) Write down the logit link $g(\pi_i)$ and inverse logit $h(\eta)$. Describe what h does geometrically.
 - (c) Suppose $\hat{\beta}_1 = 1.6$. By how much do the odds and log-odds change with a unit increase in x_1 ?
 - (d) State the score equation and its relation to cross-entropy.
 - (e) How to construct a 95% confidence interval (CI) for β_k ? What if the confidence interval contains 0?
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Question 8 — LDA: Concepts

- (a) Describe the generative approach of LDA.
 - (b) Why Gaussian assumption and shared covariance in two classes helps us to get a linear boundary?
 - (c) How do you understand the Fisher’s LDA optimization criterion?
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Question 9 — LDA: R Output & Decision Boundary

Consider the output of the LDA on slide 58.

- (a) Using slide 60, compute the midpoint $\text{mid} = (\text{mid}_1, \text{mid}_2)$.
- (b) Using slide 60, compute the equation of the decision boundary.
- (c) Why do both LDA and logistic regression produce linear decision boundaries?
- (d) When does each method outperform the other?